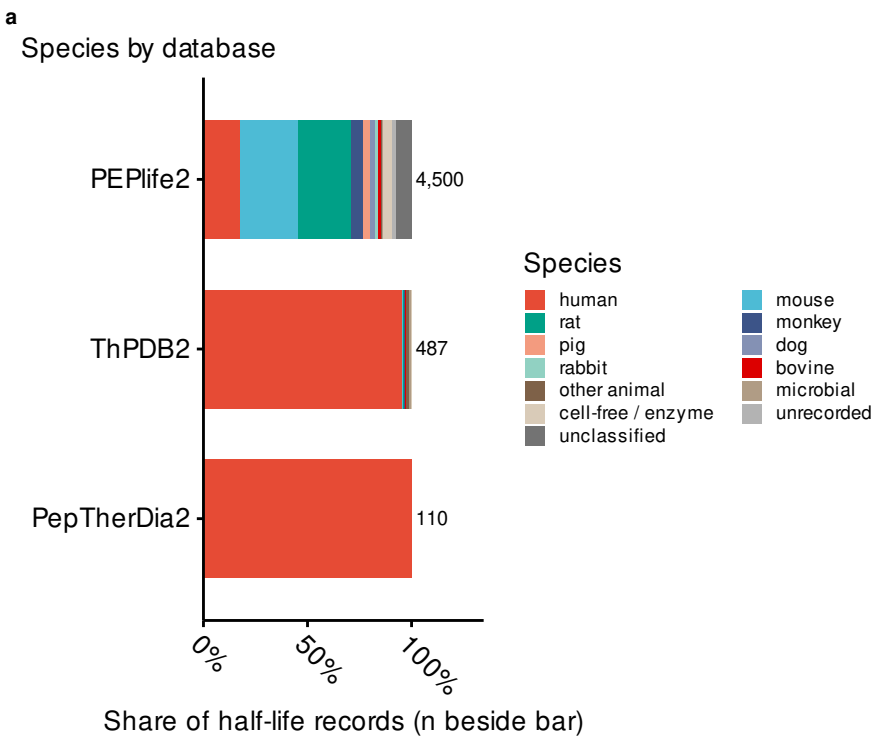
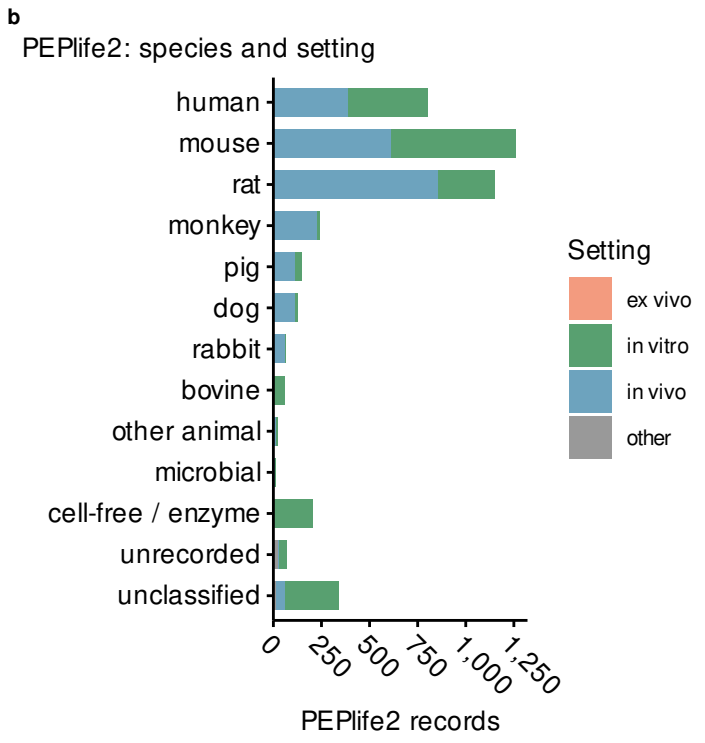
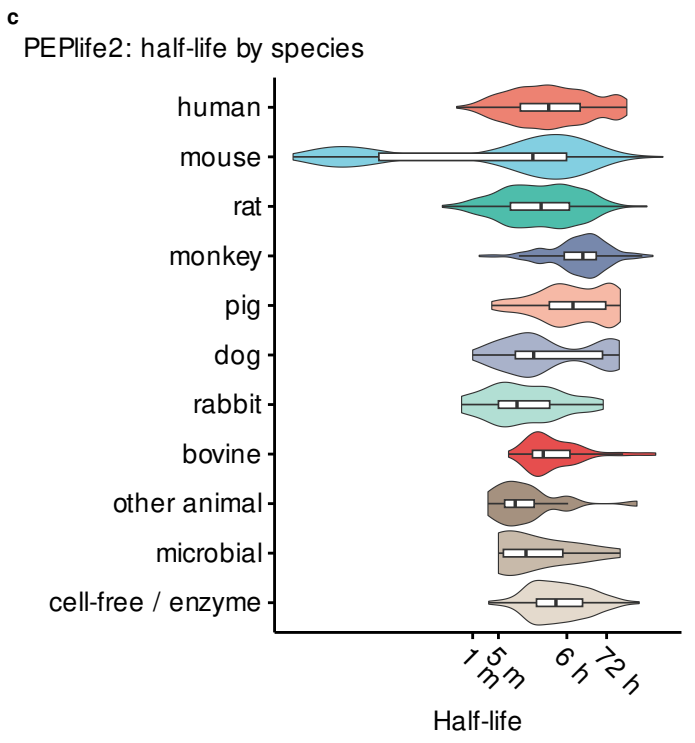
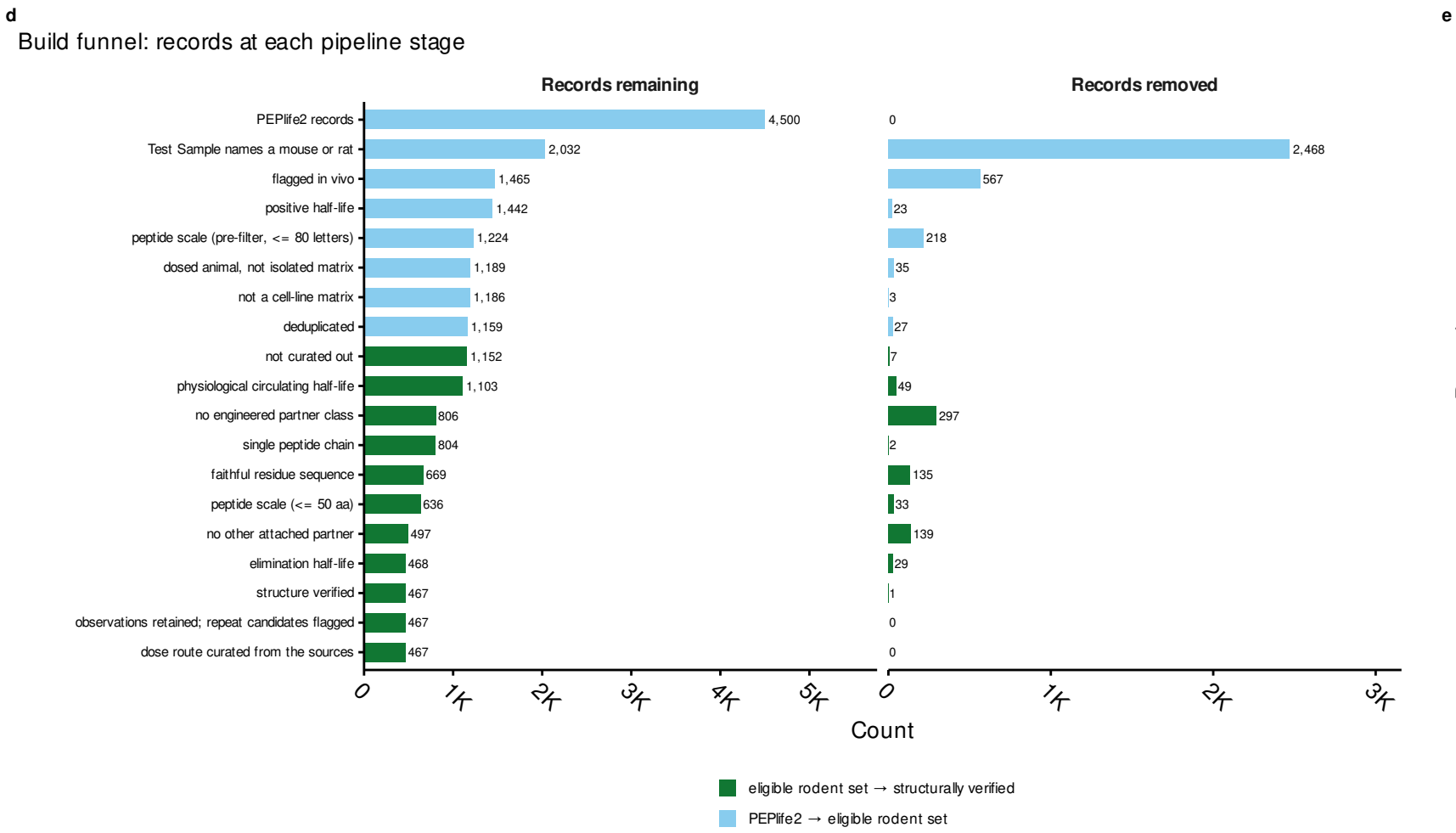
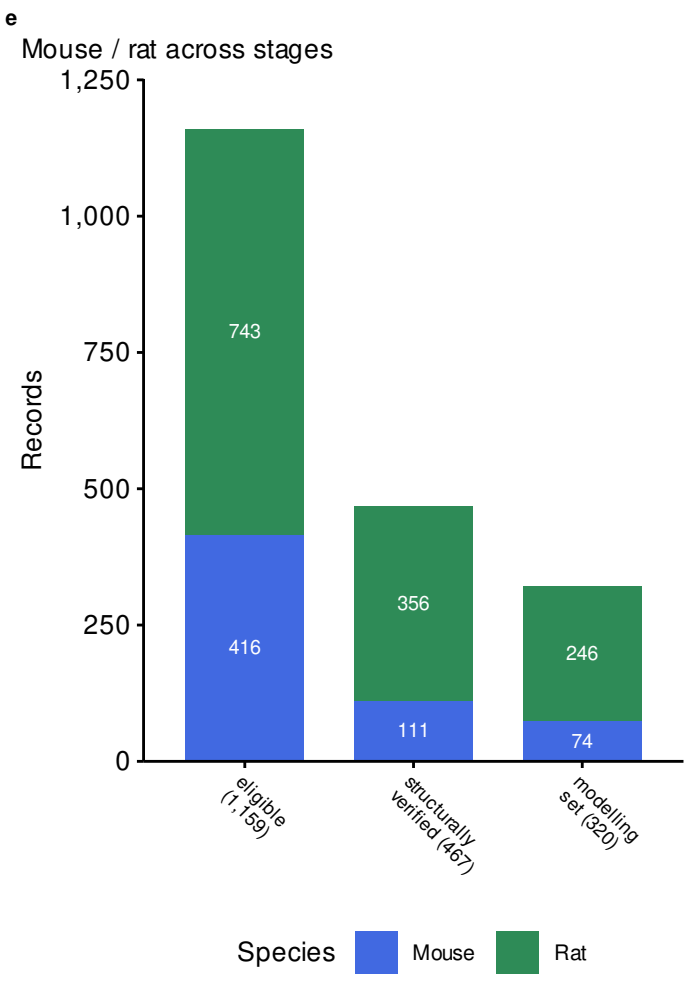
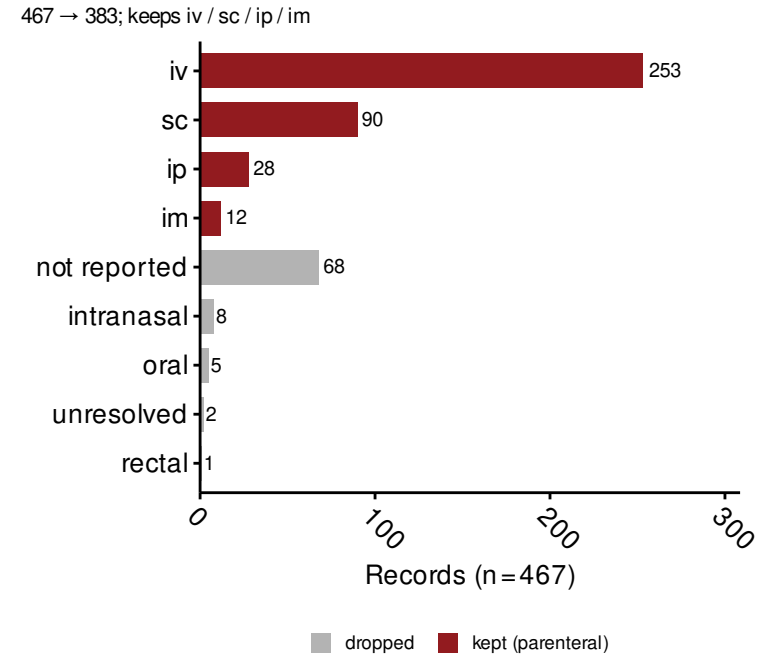

a
Species by database
PEPlife2
ThPDB2
PepTherDia2
4,500
487
110
0%
50%
100%
Share of half-life records (n beside bar)
Species
human
rat
pig
rabbit
other animal
cell-free / enzyme
unclassified
mouse
monkey
dog
bovine
microbial
unrecorded

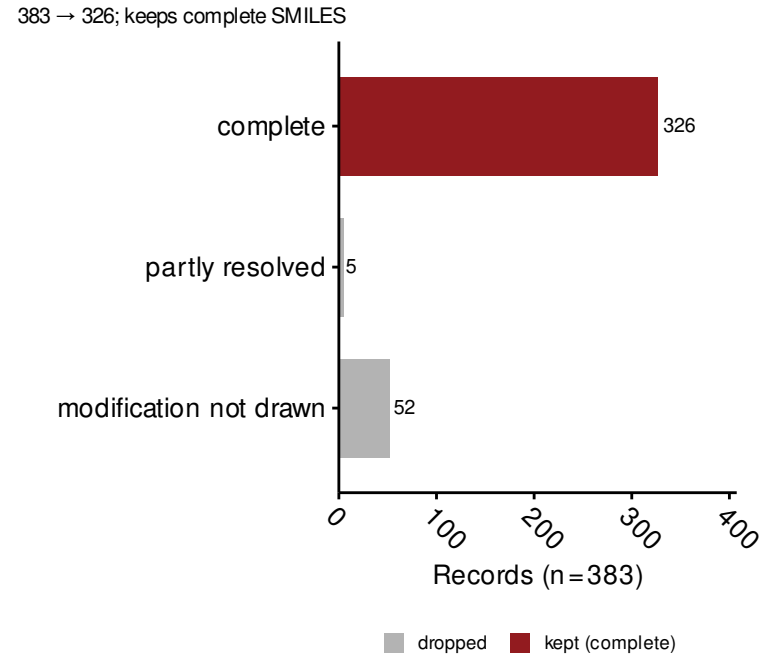

b
PEPlife2: species and setting
human
mouse
rat
monkey
pig
dog
rabbit
bovine
other animal
microbial
cell-free / enzyme
unrecorded
unclassified
0
250
500
750
1,000
1,250
PEPlife2 records
Setting
ex vivo
in vitro
in vivo
other

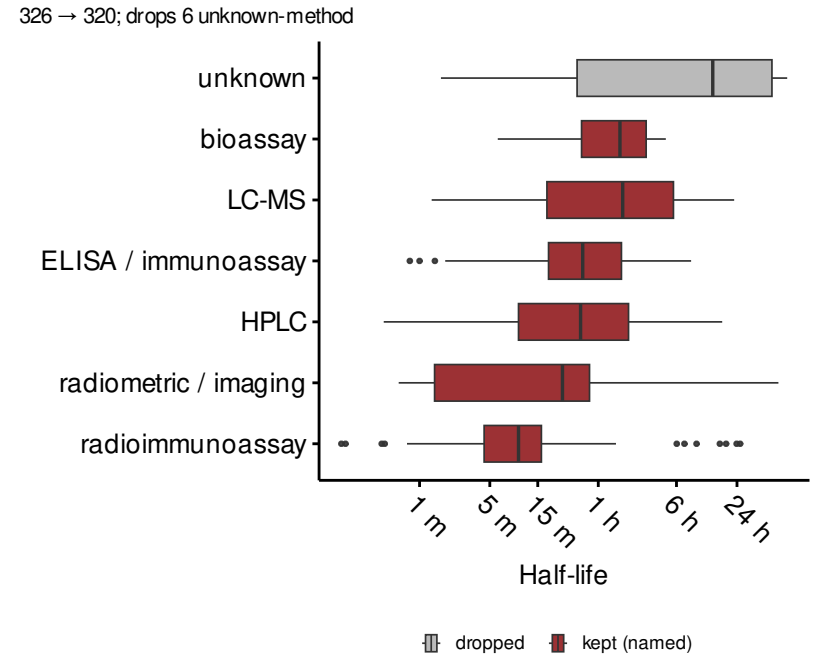

c
PEPlife2: half-life by species
human
mouse
rat
monkey
pig
dog
rabbit
bovine
other animal
microbial
cell-free / enzyme
1 m
5 m
6 h
72 h
Half-life

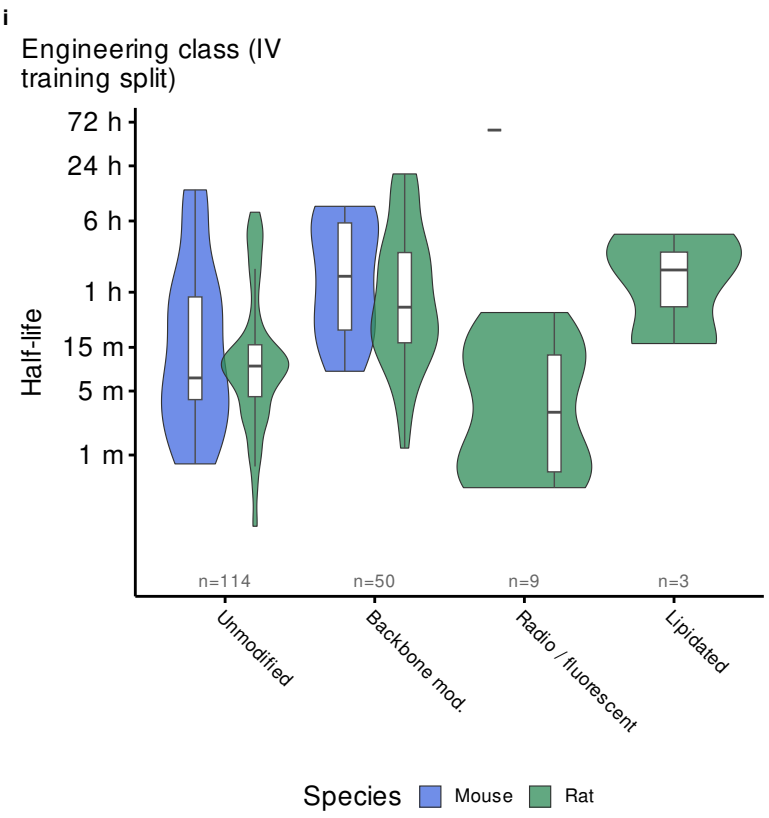

d
Build funnel: records at each pipeline stage
Records remaining
Records removed
PEPlife2 records
4,500
0
Test Sample names a mouse or rat
2,032
2,468
flagged in vivo
1,465
567
positive half-life
1,442
23
peptide scale (pre-filter, <= 80 letters)
1,224
218
dosed animal, not isolated matrix
1,189
35
not a cell-line matrix
1,186
3
deduplicated
1,159
27
not curated out
1,152
7
physiological circulating half-life
1,103
49
no engineered partner class
806
297
single peptide chain
804
2
faithful residue sequence
669
135
peptide scale (<= 50 aa)
636
33
no other attached partner
497
139
elimination half-life
468
29
structure verified
467
1
observations retained; repeat candidates flagged
467
0
dose route curated from the sources
467
0
0
1K
2K
3K
4K
5K
Count
eligible rodent set → structurally verified
PEPlife2 → eligible rodent set

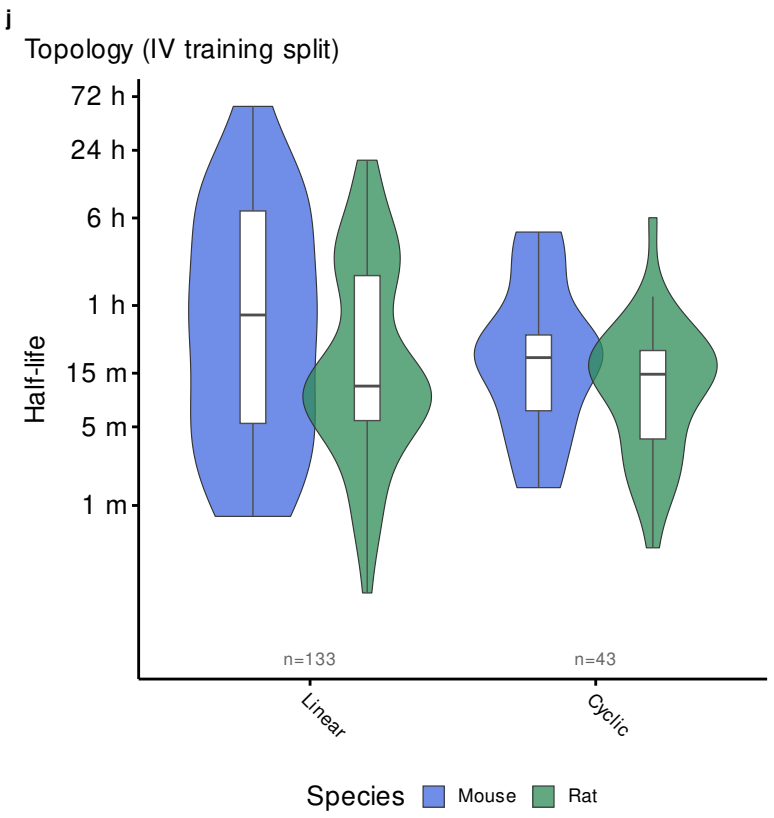

e
Mouse / rat across stages
1,250
1,000
750
500
250
0
Records
743
416
356
111
246
74
eligible (1,159)
structurally verified (467)
modelling set (320)
Species
Mouse
Rat

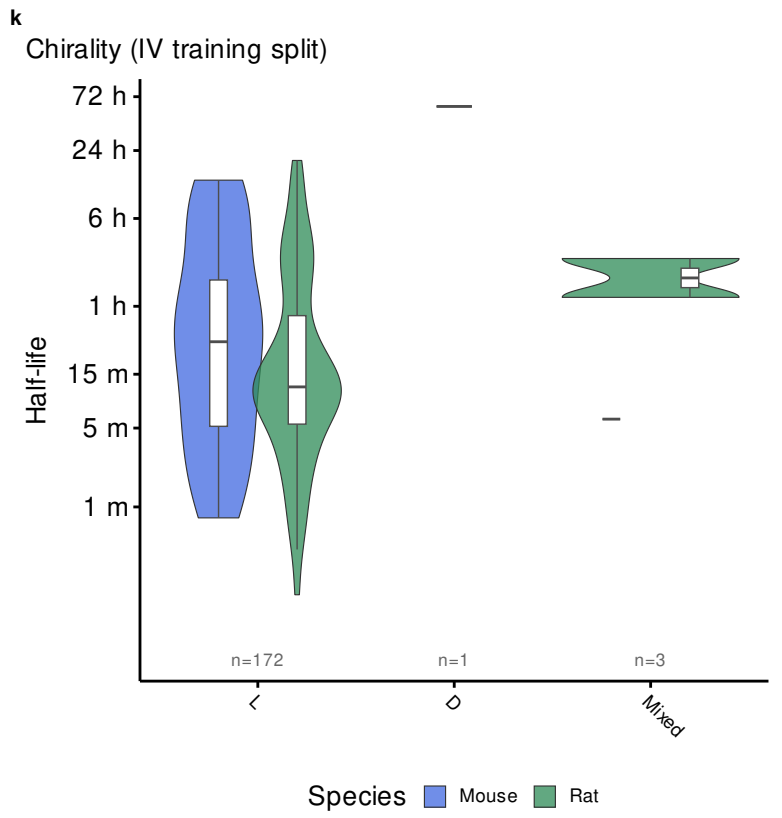

f
Route gate
467 → 383; keeps iv / sc / ip / im
iv
sc
ip
im
not reported
intranasal
oral
unresolved
rectal
253
90
28
12
68
8
5
2
1
0
100
200
300
Records (n = 467)
dropped
kept (parenteral)


g
Structure gate
383 → 326; keeps complete SMILES
complete
partly resolved
modification not drawn
326
5
52
0
100
200
300
400
Records (n = 383)
dropped
kept (complete)


h
Assay gate
326 → 320; drops 6 unknown-method
unknown
bioassay
LC-MS
ELISA / immunoassay
HPLC
radiometric / imaging
radioimmunoassay
1 m
5 m
15 m
1 h
6 h
24 h
Half-life
dropped
kept (named)


i
Engineering class (IV training split)
72 h
24 h
6 h
1 h
15 m
5 m
1 m
Half-life
n=114
n=50
n=9
n=3
Unmodified
Backbone mod.
Radio / fluorescent
Lipidated
Species
Mouse
Rat


j
Topology (IV training split)
72 h
24 h
6 h
1 h
15 m
5 m
1 m
Half-life
n=133
n=43
Linear
Cyclic
Species
Mouse
Rat


k
Chirality (IV training split)
72 h
24 h
6 h
1 h
15 m
5 m
1 m
Half-life
n=172
n=1
n=3
L
D
Mixed
Species
Mouse
Rat